

GMO IM E T KRENKELJA, Russian Federation

WMO: 200460

Lat: 80.626N

Lon: 58.059E

Elev: 81

StdP: 14.65

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	-32.2	-28.4	-36.1	0.7	-31.8	-32.9	0.9	-28.2	43.5	17.6	39.0	3.8	8.4	50	1.143

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	3.2	38.1	36.8	36.8	35.7	35.8	34.8	37.2	37.8	36.0	36.5	35.0	35.5	9.7	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
36.3	31.6	37.4	35.2	30.1	36.2	34.2	29.0	35.3	14.0	38.0	13.5	36.7	13.0	35.7	42.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
36.1	31.3	27.4	-29.7	40.6	7.9	3.1	-35.4	42.9	-40.0	44.7	-44.5	46.4	-50.2	48.7		
			-31.0	39.2	8.2	2.0	-36.9	40.6	-41.7	41.8	-46.3	42.9	-52.2	44.3		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	13.9	-1.5	-2.2	-4.5	2.0	17.0	29.7	33.3	32.6	29.4	20.6	8.8	0.3
	DBStd	17.78	15.48	15.66	13.59	11.11	7.24	3.19	1.81	2.31	3.70	9.06	13.50	14.14
	HDD50	13184	1597	1463	1689	1440	1022	609	516	540	619	912	1236	1541
	HDD65	18659	2062	1883	2154	1890	1487	1059	981	1005	1069	1377	1686	2006
	CDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
	WSAvg	12.4	13.2	12.5	13.0	12.3	11.3	11.1	10.4	10.7	12.9	14.6	13.9	13.3
	PrecAvg	10.3	1.0	1.2	0.9	0.6	0.6	0.5	0.8	0.9	1.0	0.9	1.0	1.1
(15) Precipitation	PrecMax	15.4	3.1	3.9	3.8	2.4	1.4	1.6	1.9	2.3	2.1	2.2	3.5	4.7
	PrecMin	4.2	0.1	0.0	0.0	0.0	0.1	0.0	0.1	0.2	0.2	0.2	0.0	0.0
	PrecStd	3.1	0.6	0.9	0.7	0.5	0.3	0.3	0.4	0.5	0.4	0.4	0.8	0.9
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.1	30.1	30.0	28.8	32.9	37.5	41.1	40.2	38.2	34.6	33.1	31.2
		MCWB	30.4	29.2	29.1	27.5	31.9	34.9	39.0	38.6	37.2	33.8	32.1	30.5
	2%	DB	27.8	27.8	26.8	26.3	31.0	35.4	38.3	38.0	36.5	32.9	31.3	28.2
		MCWB	26.8	27.2	26.1	25.3	29.9	33.5	36.9	36.9	35.7	32.1	30.6	27.6
	5%	DB	25.0	24.9	23.7	23.3	28.8	34.2	36.8	36.7	35.0	31.5	29.8	25.4
		MCWB	24.4	24.3	23.0	22.4	27.7	32.7	35.6	35.7	34.4	30.8	29.0	24.8
	10%	DB	21.5	21.6	18.1	18.3	26.9	33.4	35.8	35.7	34.0	30.3	27.6	21.0
		MCWB	20.7	20.9	17.5	17.5	25.9	32.1	34.7	34.8	33.4	29.5	26.8	20.4
	0.4%	WB	30.5	29.5	29.4	27.9	32.2	35.5	39.4	38.9	37.5	33.9	32.3	30.8
		MCDB	30.9	29.8	30.1	28.7	32.6	37.1	40.7	39.7	37.9	34.1	32.7	31.2
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	26.9	27.3	26.2	25.2	30.1	34.0	37.3	37.1	35.9	32.3	30.7	27.7
		MCDB	27.5	27.7	26.6	26.1	31.0	34.8	38.0	37.7	36.3	32.7	31.2	28.1
	5%	WB	24.3	24.3	23.2	22.4	27.9	33.1	35.9	35.9	34.6	31.0	29.1	24.8
		MCDB	25.0	24.9	23.9	23.3	28.7	33.9	36.5	36.4	34.9	31.5	29.7	25.4
	10%	WB	20.8	20.9	17.4	17.5	26.0	32.4	35.0	35.0	33.5	29.6	27.0	20.5
		MCDB	21.6	21.6	18.0	18.2	26.8	33.2	35.6	35.4	33.8	30.2	27.8	21.1
(35) Mean Daily Temperature Range	5% DB	MDBR	10.3	10.8	10.0	8.6	6.2	3.7	3.2	3.2	3.8	5.3	8.3	10.2
		MCDBR	11.5	14.4	13.7	11.1	7.1	4.2	4.7	4.4	4.2	5.4	6.7	11.1
		MCWBR	11.3	14.3	13.5	11.1	6.7	3.6	3.9	4.0	4.2	5.6	6.8	11.0
	5% WB	MCDBR	11.5	14.8	13.4	11.0	7.3	3.9	4.6	4.3	4.1	5.6	6.1	11.1
		MCWBR	11.4	14.7	13.2	11.0	7.0	3.5	3.9	4.0	4.2	5.8	6.4	11.0
(40) Clear-Sky Solar Irradiance	taub	N/A	N/A	0.237	0.286	0.311	0.302	0.309	0.301	0.238	N/A	N/A	N/A	N/A
	taud	N/A	N/A	1.941	2.040	2.080	2.218	2.328	2.358	2.190	N/A	N/A	N/A	N/A
	Ebn at Noon	0	0	144	224	247	260	252	228	163	0	0	0	0
	Edh at Noon	0	0	14	28	34	31	26	21	11	0	0	0	0
(44) All-Sky Solar Radiation	RadAvg	0	0	211	1023	1764	1969	1623	880	268	0	0	0	0
	RadStd	0	0	16	67	116	138	126	67	23	0	0	0	0

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page